

Batch: B1

Roll No.: 16010122158

Experiment No. 3

Title: Implement spatial data analysis in QGIS

Course Outcome:

CO2 Apply the data analytics in the field of geospatial system

Books/ Journals/ Websites referred:

QGIS Version 3.38-Vector Data Code File-World.shp

Resources used:

<https://www.qgis.org/>

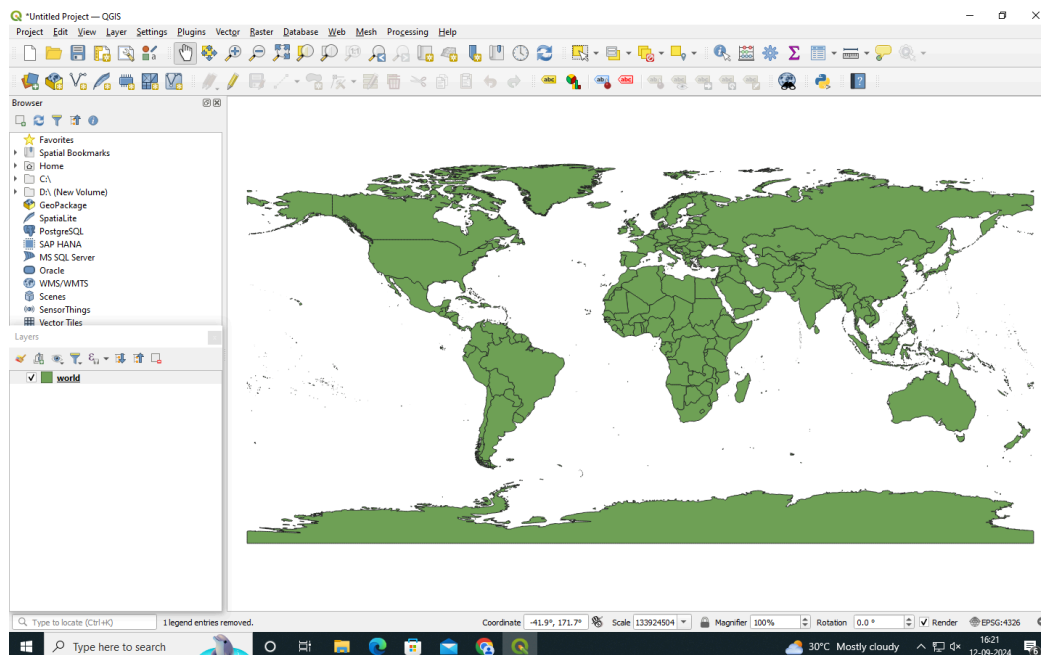
Algorithm: Spatial Data Analysis

Spatial Data type: Vector Data

Step 1: Load Your Vector Data

Open QGIS.

Add your vector layer: Go to Layer > Add Layer > Add Vector Layer... and browse to your shapefile or other vector data.



Step 2: Open the Attribute Table

Right-click on the layer in the Layers panel.

Select Open Attribute Table.

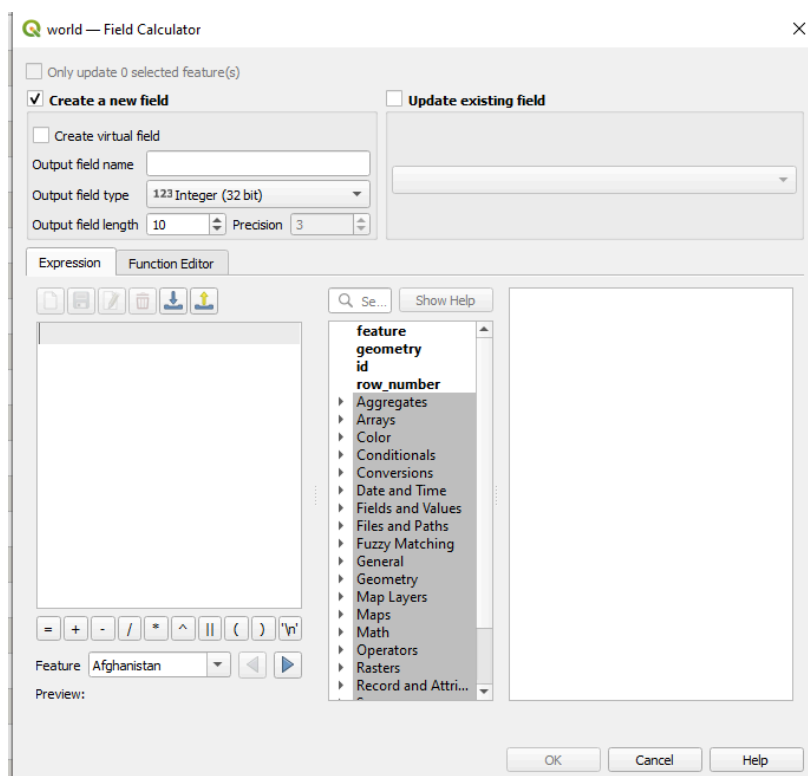
world — Features Total: 254, Filtered: 254, Selected: 0

	scalerank	featurecla	LABELRANK	SOVEREIGNT	SOV_A3	ADM0_DIF	LEVEL	TYPE	ADMIN	ADM0_A3	GEOU_DIF	GEOUNIT	GU_A3	SU_DIF
1	3	Admin-0 country	5.00	Netherlands	NL1	1.00	2.00	Country	Aruba	ABW	0	Aruba	ABW	0
2	0	Admin-0 country	3.00	Afghanistan	AFG	0	2.00	Sovereign coun...	Afghanistan	AFG	0	Afghanistan	AFG	0
3	0	Admin-0 country	3.00	Angola	AGO	0	2.00	Sovereign coun...	Angola	AGO	0	Angola	AGO	0
4	3	Admin-0 country	6.00	United Kingdom	GB1	1.00	2.00	Dependency	Anguilla	AIA	0	Anguilla	AIA	0
5	0	Admin-0 country	6.00	Albania	ALB	0	2.00	Sovereign coun...	Albania	ALB	0	Albania	ALB	0
6	3	Admin-0 country	6.00	Finland	FI1	1.00	2.00	Country	Aland	ALD	0	Aland	ALD	0
7	0	Admin-0 country	6.00	Andorra	AND	0	2.00	Sovereign coun...	Andorra	AND	0	Andorra	AND	0
8	0	Admin-0 country	4.00	United Arab Em...	ARE	0	2.00	Sovereign coun...	United Arab Em...	ARE	0	United Arab Em...	ARE	0
9	0	Admin-0 country	2.00	Argentina	ARG	0	2.00	Sovereign coun...	Argentina	ARG	0	Argentina	ARG	0
10	0	Admin-0 country	6.00	Armenia	ARM	0	2.00	Sovereign coun...	Armenia	ARM	0	Armenia	ARM	0
11	5	Admin-0 country	4.00	United States of...	US1	1.00	2.00	Dependency	American Samoa	ASM	0	American Samoa	ASM	0
12	0	Admin-0 country	4.00	Antarctica	ATA	0	2.00	Indeterminate	Antarctica	ATA	0	Antarctica	ATA	0
13	6	Admin-0 country	5.00	Australia	AU1	1.00	2.00	Dependency	Ashmore and C...	ATC	0	Ashmore and C...	ATC	0
14	3	Admin-0 country	6.00	France	FR1	1.00	2.00	Dependency	French Souther...	ATF	0	French Souther...	ATF	0
15	3	Admin-0 country	6.00	Antigua and Ba...	ATG	0	2.00	Sovereign coun...	Antigua and Ba...	ATG	0	Antigua and Ba...	ATG	0
16	0	Admin-0 country	2.00	Australia	AU1	1.00	2.00	Country	Australia	AUS	0	Australia	AUS	0
17	0	Admin-0 country	4.00	Austria	AUT	0	2.00	Sovereign coun...	Austria	AUT	0	Austria	AUT	0
18	0	Admin-0 country	5.00	Azerbaijan	AZE	0	2.00	Sovereign coun...	Azerbaijan	AZE	0	Azerbaijan	AZE	0
19	0	Admin-0 country	6.00	Burundi	BDI	0	2.00	Sovereign coun...	Burundi	BDI	0	Burundi	BDI	0
20	0	Admin-0 country	2.00	Belgium	BEL	0	2.00	Sovereign coun...	Belgium	BEL	0	Belgium	BEL	0
21	0	Admin-0 country	5.00	Benin	BEN	0	2.00	Sovereign coun...	Benin	BEN	0	Benin	BEN	0
22	0	Admin-0 country	3.00	Burkina Faso	BFA	0	2.00	Sovereign coun...	Burkina Faso	BFA	0	Burkina Faso	BFA	0
23	0	Admin-0 country	3.00	Bangladesh	BGD	0	2.00	Sovereign coun...	Bangladesh	BGD	0	Bangladesh	BGD	0
24	0	Admin-0 country	4.00	Bulgaria	BGR	0	2.00	Sovereign coun...	Bulgaria	BGR	0	Bulgaria	BGR	0
25	5	Admin-0 country	4.00	Bahrain	BHR	0	2.00	Sovereign coun...	Bahrain	BHR	0	Bahrain	BHR	0

Show All Features

Step 3: Add a New Field for Calculations

In the attribute table, click on the Field Calculator icon (it looks like an abacus).



Step 4: Calculate Area

In the Field Calculator dialog:

Check the option Create a new field.

Enter a name for the new field (e.g., "Area").

Set the output field type to Decimal number (real).

In the Expression field, enter the following expression to calculate the area in square meters:

\$area

	ADM0_A3_WB	CONTINENT	REGION_UN	SUBREGION	REGION_WB	NAME_LEN	LONG_LEN	ABBREV_LEN	TINY	HOMEPART	MIN_ZOOM	MIN_LABEL	MAX_LABEL	AREA
1	-99.00	North America	Americas	Caribbean	Latin America ...	5.00	5.00	5.00	4.00	-99.00	0	5.00	10.00	169766422.380
2	-99.00	Asia	Asia	Southern Asia	South Asia	11.00	11.00	4.00	-99.00	1.00	0	3.00	7.00	642181644968.554
3	-99.00	Africa	Africa	Middle Africa	Sub-Saharan Af...	6.00	6.00	4.00	-99.00	1.00	0	3.00	7.00	1244654322797...
4	-99.00	North America	Americas	Caribbean	Latin America ...	8.00	8.00	4.00	-99.00	-99.00	0	5.00	10.00	80673845.974
5	-99.00	Europe	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	28335848832.045
6	-99.00	Europe	Europe	Northern Europe	Europe & Centr...	5.00	13.00	5.00	5.00	-99.00	0	5.00	10.00	942053624.128
7	-99.00	Europe	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	5.00	1.00	0	5.00	10.00	452247936.840
8	-99.00	Asia	Asia	Western Asia	Middle East & ...	20.00	20.00	6.00	-99.00	1.00	0	4.00	9.00	71084655105.556
9	-99.00	South America	Americas	South America	Latin America ...	9.00	9.00	4.00	-99.00	1.00	0	2.00	7.00	2784305848224...
10	-99.00	Asia	Asia	Western Asia	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	29588310318.547
11	-99.00	Oceania	Oceania	Polynesia	East Asia & Pac...	14.00	14.00	9.00	3.00	-99.00	0	4.00	9.00	180304292.357
12	-99.00	Antarctica	Antarctica	Antarctica	Antarctica	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	1235817436893...
13	-99.00	Oceania	Oceania	Australia and N...	East Asia & Pac...	23.00	27.00	7.00	-99.00	-99.00	0	4.50	9.50	2695492.866
14	-99.00	Seven seas (ope...	Seven seas (ope...	Sub-Saharan Af...	Sub-Saharan Af...	22.00	35.00	10.00	2.00	-99.00	0	4.00	9.00	72799857996.489
15	-99.00	North America	Americas	Caribbean	Latin America ...	17.00	19.00	6.00	4.00	1.00	0	5.00	9.50	451480188.791
16	-99.00	Oceania	Oceania	Australia and N...	East Asia & Pac...	9.00	9.00	4.00	-99.00	1.00	0	1.70	5.70	7691175090399...
17	-99.00	Europe	Europe	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	3.00	8.00	83993198286.501
18	-99.00	Asia	Asia	Western Asia	Europe & Centr...	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	86249000824.298
19	-99.00	Africa	Africa	Eastern Africa	Sub-Saharan Af...	7.00	7.00	4.00	-99.00	1.00	0	4.00	9.00	27041266296.112
20	-99.00	Europe	Europe	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	4.00	9.00	30669830581.700

To get the area in hectares:

\$area / 10000

Click OK to create the new field and calculate the areas.

	ADM0_A3_WB	CONTINENT	REGION_UN	SUBREGION	REGION_WB	NAME_LEN	LONG_LEN	ABBREV_LEN	TINY	HOMEPART	MIN_ZOOM	MIN_LABEL	MAX_LABEL	AREA/1000	
1	-99.00	North America	Americas	Caribbean	Latin America ...	5.00	5.00	5.00	4.00	-99.00	0	5.00	10.00	16977	
2	-99.00	Asia	Asia	Southern Asia	South Asia	11.00	11.00	4.00	-99.00	1.00	0	3.00	7.00	64218164	
3	-99.00	Africa	Africa	Middle Africa	Sub-Saharan Af...	6.00	6.00	4.00	-99.00	1.00	0	3.00	7.00	124465432	
4	-99.00	North America	Americas	Caribbean	Latin America ...	8.00	8.00	4.00	-99.00	-99.00	0	5.00	10.00	8067	
5	-99.00	Europe	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	2833585	
6	-99.00	Europe	Europe	Northern Europe	Europe & Centr...	5.00	13.00	5.00	5.00	-99.00	0	5.00	10.00	94205	
7	-99.00	Europe	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	5.00	1.00	0	5.00	10.00	45225	
8	-99.00	Asia	Asia	Western Asia	Middle East & ...	20.00	20.00	6.00	-99.00	1.00	0	4.00	9.00	7108466	
9	-99.00	South America	Americas	South America	Latin America ...	9.00	9.00	4.00	-99.00	1.00	0	2.00	7.00	278430585	
10	-99.00	Asia	Asia	Western Asia	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	2958831	
11	-99.00	Oceania	Oceania	Polynesia	East Asia & Pac...	14.00	14.00	9.00	3.00	-99.00	0	4.00	9.00	18030	
12	-99.00	Antarctica	Antarctica	Antarctica	Antarctica	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	1235817437	
13	-99.00	Oceania	Oceania	Australia and N...	East Asia & Pac...	23.00	27.00	7.00	-99.00	-99.00	0	4.50	9.50	270	
14	-99.00	Seven seas (ope...	Seven seas (ope...	Seven seas (ope...	Sub-Saharan Af...	22.00	35.00	10.00	2.00	-99.00	0	4.00	9.00	727999	
15	-99.00	North America	Americas	Caribbean	Latin America ...	17.00	35.00	19.00	6.00	4.00	1.00	0	5.00	9.50	45148
16	-99.00	Oceania	Oceania	Australia and N...	East Asia & Pac...	9.00	9.00	4.00	-99.00	1.00	0	1.70	5.70	769117509	
17	-99.00	Europe	Europe	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	3.00	8.00	8399320	
18	-99.00	Asia	Asia	Western Asia	Europe & Centr...	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	8624980	

Step : Calculate Perimeter (for polygons):

perimeter(\$geometry)

	ADM0_A3_WB	CONTINENT	REGION_UN	SUBREGION	REGION_WB	NAME_LEN	LONG_LEN	ABBREV_LEN	TINY	HOMEPART	MIN_ZOOM	MIN_LABEL	MAX_LABEL	Perimeter
1	-99.00	North America	Americas	Caribbean	Latin America ...	5.00	5.00	5.00	4.00	-99.00	0	5.00	10.00	1
2	-99.00	Asia	Asia	Southern Asia	South Asia	11.00	11.00	4.00	-99.00	1.00	0	3.00	7.00	53
3	-99.00	Africa	Africa	Middle Africa	Sub-Saharan Af...	6.00	6.00	4.00	-99.00	1.00	0	3.00	7.00	62
4	-99.00	North America	Americas	Caribbean	Latin America ...	8.00	8.00	4.00	-99.00	-99.00	0	5.00	10.00	1
5	-99.00	Europe	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	11
6	-99.00	Europe	Europe	Northern Europe	Europe & Centr...	5.00	13.00	5.00	5.00	-99.00	0	5.00	10.00	8
7	-99.00	Europe	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	5.00	1.00	0	5.00	10.00	1
8	-99.00	Asia	Asia	Western Asia	Middle East & ...	20.00	20.00	6.00	-99.00	1.00	0	4.00	9.00	21
9	-99.00	South America	Americas	South America	Latin America ...	9.00	9.00	4.00	-99.00	1.00	0	2.00	7.00	152
10	-99.00	Asia	Asia	Western Asia	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	13
11	-99.00	Oceania	Oceania	Polynesia	East Asia & Pac...	14.00	14.00	9.00	3.00	-99.00	0	4.00	9.00	1
12	-99.00	Antarctica	Antarctica	Antarctica	Antarctica	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	1952
13	-99.00	Oceania	Oceania	Australia and N...	East Asia & Pac...	23.00	27.00	7.00	-99.00	-99.00	0	4.50	9.50	0
14	-99.00	Seven seas (ope...	Seven seas (ope...	Seven seas (ope...	Sub-Saharan Af...	22.00	35.00	10.00	2.00	-99.00	0	4.00	9.00	18
15	-99.00	North America	Americas	Caribbean	Latin America ...	17.00	19.00	6.00	4.00	1.00	0	5.00	9.50	2
16	-99.00	Oceania	Oceania	Australia and N...	East Asia & Pac...	9.00	9.00	4.00	-99.00	1.00	0	1.70	5.70	323
17	-99.00	Europe	Europe	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	3.00	8.00	24
18	-99.00	Asia	Asia	Western Asia	Europe & Centr...	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	28
19	-99.00	Africa	Africa	Eastern Africa	Sub-Saharan Af...	7.00	7.00	4.00	-99.00	1.00	0	4.00	9.00	8

Step: length of all countries

Task: To work on other geometric properties like centroid using different expressions in the Field Calculator.

Centroid_x:

	CONTINENT	REGION_UN	SUBREGION	REGION_WB	NAME_LEN	LONG_LEN	ABBREV_LEN	TINY	HOMEPART	MIN_ZOOM	MIN_LABEL	MAX_LABEL	length	centroid_x
1	North America	Americas	Caribbean	Latin America ...	5.00	5.00	5.00	4.00	-99.00	0	5.00	10.00	NULL	-69.974
2	Asia	Asia	Southern Asia	South Asia	11.00	11.00	4.00	-99.00	1.00	0	3.00	7.00	NULL	66.008
3	Africa	Africa	Middle Africa	Sub-Saharan Af...	6.00	6.00	4.00	-99.00	1.00	0	3.00	7.00	NULL	-17.537
4	North America	Americas	Caribbean	Latin America ...	8.00	8.00	4.00	-99.00	-99.00	0	5.00	10.00	NULL	-63.059
5	Europe	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	NULL	20.054
6	Europe	Europe	Northern Europe	Europe & Centr...	5.00	13.00	5.00	5.00	-99.00	0	5.00	10.00	NULL	20.056
7	Europe	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	5.00	1.00	0	5.00	10.00	NULL	1.561
8	Asia	Asia	Western Asia	Middle East & ...	20.00	20.00	6.00	-99.00	1.00	0	4.00	9.00	NULL	54.301
9	South America	Americas	South America	Latin America ...	9.00	9.00	4.00	-99.00	1.00	0	2.00	7.00	NULL	-65.178
10	Asia	Asia	Western Asia	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	NULL	44.932
11	Oceania	Oceania	Polynesia	East Asia & Pac...	14.00	14.00	9.00	3.00	-99.00	0	4.00	9.00	NULL	-170.399
12	Antarctica	Antarctica	Antarctica	Antarctica	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	19.908
13	Oceania	Oceania	Australia and N...	East Asia & Pac...	23.00	27.00	7.00	-99.00	-99.00	0	4.50	9.50	NULL	123.586
14	Seven seas (ope...	Seven seas (ope...	Seven seas (ope...	Sub-Saharan Af...	22.00	35.00	10.00	2.00	-99.00	0	4.00	9.00	NULL	68.992
15	North America	Americas	Caribbean	Latin America ...	17.00	19.00	6.00	4.00	1.00	0	5.00	9.50	NULL	-61.792
16	Oceania	Oceania	Australia and N...	East Asia & Pac...	9.00	9.00	4.00	-99.00	1.00	0	1.70	5.70	NULL	134.491
17	Europe	Europe	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	3.00	8.00	NULL	14.131
18	Asia	Asia	Western Asia	Europe & Centr...	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	47.542
19	Africa	Africa	Eastern Africa	Sub-Saharan Af...	7.00	7.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	29.875

Centroid_y

	REGION_UN	SUBREGION	REGION_WB	NAME_LEN	LONG_LEN	ABBREV_LEN	TINY	HOMEPART	MIN_ZOOM	MIN_LABEL	MAX_LABEL	length	centroid_x	Cebtrid_y
1	Americas	Caribbean	Latin America ...	5.00	5.00	5.00	4.00	-99.00	0	5.00	10.00	NULL	-69.974	12.517
2	Asia	Southern Asia	South Asia	11.00	11.00	4.00	-99.00	1.00	0	3.00	7.00	NULL	66.008	33.836
3	Africa	Middle Africa	Sub-Saharan Af...	6.00	6.00	4.00	-99.00	1.00	0	3.00	7.00	NULL	17.537	-12.292
4	Americas	Caribbean	Latin America ...	8.00	8.00	4.00	-99.00	-99.00	0	5.00	10.00	NULL	-63.059	18.226
5	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	NULL	20.054	41.142
5	Europe	Northern Europe	Europe & Centr...	5.00	13.00	5.00	5.00	-99.00	0	5.00	10.00	NULL	20.056	60.186
7	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	5.00	1.00	0	5.00	10.00	NULL	1.561	42.541
3	Asia	Western Asia	Middle East & ...	20.00	20.00	6.00	-99.00	1.00	0	4.00	9.00	NULL	54.301	23.906
9	Americas	South America	Latin America ...	9.00	9.00	4.00	-99.00	1.00	0	2.00	7.00	NULL	-65.178	-35.388
10	Asia	Western Asia	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	NULL	44.932	40.288
11	Oceania	Polynesia	East Asia & Pac...	14.00	14.00	9.00	3.00	-99.00	0	4.00	9.00	NULL	-170.399	-14.255
12	Antarctica	Antarctica	Antarctica	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	19.908	-80.508
13	Oceania	Australia and N...	East Asia & Pac...	23.00	27.00	7.00	-99.00	-99.00	0	4.50	9.50	NULL	123.586	-12.433
14	Seven seas (ope...	Seven seas (ope...	Sub-Saharan Af...	22.00	35.00	10.00	2.00	-99.00	0	4.00	9.00	NULL	68.992	-49.065
15	Americas	Caribbean	Latin America ...	17.00	19.00	6.00	4.00	1.00	0	5.00	9.50	NULL	-61.792	17.284
16	Oceania	Australia and N...	East Asia & Pac...	9.00	9.00	4.00	-99.00	1.00	0	1.70	5.70	NULL	134.491	-25.732
17	Europe	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	3.00	8.00	NULL	14.131	47.586
18	Asia	Western Asia	Europe & Centr...	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	47.542	40.289
19	Africa	Eastern Africa	Sub-Saharan Af...	7.00	7.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	29.875	-3.357

Centroid Distance

	REGION_UN	SUBREGION	REGION_WB	NAME_LEN	LONG_LEN	ABBREV_LEN	TINY	HOMEPART	MIN_ZOOM	MIN_LABEL	MAX_LABEL	length	centroid_x	cent_dist
1	Americas	Caribbean	Latin America ...	5.00	5.00	5.00	4.00	-99.00	0	5.00	10.00	NULL	-69.974	0
2	Asia	Southern Asia	South Asia	11.00	11.00	4.00	-99.00	1.00	0	3.00	7.00	NULL	66.008	0
3	Africa	Middle Africa	Sub-Saharan Af...	6.00	6.00	4.00	-99.00	1.00	0	3.00	7.00	NULL	17.537	0
4	Americas	Caribbean	Latin America ...	8.00	8.00	4.00	-99.00	-99.00	0	5.00	10.00	NULL	-63.059	0
5	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	NULL	20.054	0
6	Europe	Northern Europe	Europe & Centr...	5.00	13.00	5.00	5.00	-99.00	0	5.00	10.00	NULL	20.056	0.005
7	Europe	Southern Europe	Europe & Centr...	7.00	7.00	4.00	5.00	1.00	0	5.00	10.00	NULL	1.561	0
8	Asia	Western Asia	Middle East & ...	20.00	20.00	6.00	-99.00	1.00	0	4.00	9.00	NULL	54.301	0
9	Americas	South America	Latin America ...	9.00	9.00	4.00	-99.00	1.00	0	2.00	7.00	NULL	-65.178	0
10	Asia	Western Asia	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	NULL	44.932	0
11	Oceania	Polynesia	East Asia & Pac...	14.00	14.00	9.00	3.00	-99.00	0	4.00	9.00	NULL	-170.399	0.168
12	Antarctica	Antarctica	Antarctica	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	19.908	0
13	Oceania	Australia and N...	East Asia & Pac...	23.00	27.00	7.00	-99.00	-99.00	0	4.50	9.50	NULL	123.586	0
14	Seven seas (ope...	Seven seas (ope...	Sub-Saharan Af...	22.00	35.00	10.00	2.00	-99.00	0	4.00	9.00	NULL	68.992	0
15	Americas	Caribbean	Latin America ...	17.00	19.00	6.00	4.00	1.00	0	5.00	9.50	NULL	-61.792	0.119
16	Oceania	Australia and N...	East Asia & Pac...	9.00	9.00	4.00	-99.00	1.00	0	1.70	5.70	NULL	134.491	0
17	Europe	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	3.00	8.00	NULL	14.131	0

Intersecting Areas

	SUBREGION	REGION_WB	NAME_LEN	LONG_LEN	ABBREV_LEN	TINY	HOMEPART	MIN_ZOOM	MIN_LABEL	MAX_LABEL	length	centroid_x	cent_dist	intst_area
1	Caribbean	Latin America ...	5.00	5.00	5.00	4.00	-99.00	0	5.00	10.00	NULL	-69.974	0	0.014
2	Southern Asia	South Asia	11.00	11.00	4.00	-99.00	1.00	0	3.00	7.00	NULL	66.008	0	62.592
3	Middle Africa	Sub-Saharan Af...	6.00	6.00	4.00	-99.00	1.00	0	3.00	7.00	NULL	17.537	0	103.585
4	Caribbean	Latin America ...	8.00	8.00	4.00	-99.00	-99.00	0	5.00	10.00	NULL	-63.059	0	0.007
5	Southern Europe	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	NULL	20.054	0	3.039
6	Northern Europe	Europe & Centr...	5.00	13.00	5.00	5.00	-99.00	0	5.00	10.00	NULL	20.056	0.005	0.152
7	Southern Europe	Europe & Centr...	7.00	7.00	4.00	5.00	1.00	0	5.00	10.00	NULL	1.561	0	0.05
8	Western Asia	Middle East & ...	20.00	20.00	6.00	-99.00	1.00	0	4.00	9.00	NULL	54.301	0	6.304
9	South America	Latin America ...	9.00	9.00	4.00	-99.00	1.00	0	2.00	7.00	NULL	-65.178	0	278.681
10	Western Asia	Europe & Centr...	7.00	7.00	4.00	-99.00	1.00	0	5.00	10.00	NULL	44.932	0	3.134
11	Polynesia	East Asia & Pac...	14.00	14.00	9.00	3.00	-99.00	0	4.00	9.00	NULL	-170.399	0.168	0.015
12	Antarctica	Antarctica	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	19.908	0	6049.575
13	Australia and N...	East Asia & Pac...	23.00	27.00	7.00	-99.00	-99.00	0	4.50	9.50	NULL	123.586	0	0
14	Seven seas (ope...	Sub-Saharan Af...	22.00	35.00	10.00	2.00	-99.00	0	4.00	9.00	NULL	68.992	0	0.896
15	Caribbean	Latin America ...	17.00	19.00	6.00	4.00	1.00	0	5.00	9.50	NULL	-61.792	0.119	0.038
16	Australia and N...	East Asia & Pac...	9.00	9.00	4.00	-99.00	1.00	0	1.70	5.70	NULL	134.491	0	695.909
17	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	3.00	8.00	NULL	14.131	0	10.044
18	Western Asia	Europe & Centr...	10.00	10.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	47.542	0	9.135
19	Eastern Africa	Sub-Saharan Af...	7.00	7.00	4.00	-99.00	1.00	0	4.00	9.00	NULL	29.875	0	2.201
20	Western Europe	Europe & Centr...	7.00	7.00	5.00	-99.00	1.00	0	4.00	9.00	NULL	4.640	0	3.897
21	Western Africa	Sub-Saharan Af...	5.00	5.00	5.00	-99.00	1.00	0	4.00	9.00	NULL	2.329	0	9.568

Following points should be written by students

Different Geometric Properties used in spatial data analysis.

Geometric properties in spatial data analysis refer to the measurable attributes of vector data, such as points, lines, and polygons, that help define spatial features. Key properties include area, which measures the surface covered by a polygon, and perimeter, which calculates the boundary length of a polygon. For linear features, the length property is used to determine the total distance covered by lines or boundaries. Additionally, centroid coordinates (Centroid_x and Centroid_y) help identify the geometric center of a feature, aiding in the spatial alignment or distribution of objects.

Another important aspect is the calculation of distances between centroids, which helps in understanding spatial relationships between different features. In some cases, intersecting areas can be determined to analyze where two or more features overlap. These properties are critical for tasks like land use planning, environmental monitoring, and infrastructure development, making spatial data analysis an essential tool in fields like urban development, agriculture, and resource management.

Conclusion:

We used QGIS to calculate geometric properties like area, perimeter, and centroid coordinates on vector data. This demonstrated the software's ability to perform spatial operations efficiently, laying the groundwork for more advanced geospatial analysis.

Post lab questions:

Q.1 What are different geometric properties used in spatial data analysis on vector data with expression?

Area: `$area`

Perimeter: `perimeter($geometry)`

Length: `length($geometry)`

Centroid (X-coordinate): `x(centroid($geometry))`

Centroid (Y-coordinate): `y(centroid($geometry))`

Distance between centroids: `distance(centroid($geometry), centroid($other_geometry))`

Intersecting areas: `intersects($geometry, $other_geometry)`

Q.2 What are different geometric properties used in spatial data analysis on raster data with expression?

Pixel value: Represents the value of the phenomenon the raster captures, such as elevation, temperature, or intensity.

Slope: `slope()` (calculated from elevation data).

Aspect: `aspect()` (the direction of the slope).

Raster statistics: Calculations like mean, sum, or standard deviation of pixel values over a specified area.

Q.3 What is spatial data analysis and its advantages

Spatial data analysis refers to the process of examining the locations, attributes, and relationships of features in spatial data through the use of statistical and geographic techniques. It helps in identifying patterns, relationships, and trends within the data.

Advantages:

- Helps in decision-making by analyzing the spatial distribution of phenomena.
- Improves resource management (e.g., land use planning, urban development).
- Enhances environmental monitoring (e.g., deforestation, pollution).
- Facilitates real-time data tracking and analysis (e.g., for disaster management, infrastructure development).